

Delicate Courier

Route Optimizer

Technical Documentation & ROI Analysis
Last-Mile Delivery Route Planning System

Pretoria / Tshwane Metropolitan Area
3-Driver Fleet Operations
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Table of Contents

- 1. Executive Summary**
- 2. Problem Statement**
- 3. Solution Overview**
- 4. System Architecture & Tech Stack**
- 5. Feature Catalogue**
 - 5.1 Dispatcher Dashboard
 - 5.2 Route Optimization Engine
 - 5.3 Driver Companion App (PWA)
 - 5.4 Live Tracking & Traffic
 - 5.5 Data Centre & Archival
 - 5.6 Insights Engine
- 6. API Cost Optimization**
- 7. Security & Authentication**
- 8. Database Architecture**
- 9. ROI & Cost-Benefit Analysis**
- 10. Project Statistics**

1. Executive Summary

The Delicate Courier Route Optimizer is a comprehensive, production-grade last-mile delivery management platform built specifically for Delicate Courier's operations in the Pretoria/Tshwane metropolitan area. The system replaces manual route planning with an intelligent, automated solution that optimizes delivery routes for a 3-driver fleet, integrates real-time traffic data, and provides complete operational visibility from dispatch to proof of delivery.

The platform comprises two interconnected applications: a full-featured dispatcher dashboard for operations management and a Progressive Web App (PWA) for drivers in the field. Together, they digitize the entire delivery lifecycle — from CSV shipment import through route optimization, live tracking, stop completion with signature capture, and historical data archival.

Key business outcomes include an estimated 25-35% reduction in daily driving distance through algorithmic route optimization, 65-75% reduction in Google API costs through aggressive caching and smart request management, and elimination of manual route planning effort — previously consuming 45-60 minutes daily.

2. Problem Statement

Before This System

- Route planning was done manually by the dispatcher each morning, taking 45-60 minutes
- No visibility into driver locations once they left the depot
 - No traffic awareness — drivers often hit congestion on N1/N4/N14 corridors without warning
 - Shipment data from ShipLogic had to be manually interpreted and assigned to drivers
 - No historical record of deliveries — making performance analysis impossible
 - Driver communication relied on phone calls and WhatsApp messages
 - No proof of delivery system — disputes had no digital evidence trail
 - Vehicle constraints (Vitz capacity, bread consignment rules) tracked mentally
 - Saturday delivery caps not systematically enforced
 - No fuel cost tracking or route efficiency metrics

Core Challenges

- Pretoria's complex road network with frequent construction and road closures
 - Mix of collections and deliveries requiring careful sequencing
 - Express/priority shipments needing special handling within time windows
 - Multiple depots across the metro with different driver shift patterns
 - Duplicate suburb names across

Pretoria requiring geo-disambiguation

3. Solution Overview

The Delicate Courier Route Optimizer provides an end-to-end delivery management solution with the following core capabilities:

ShipLogic CSV Import

One-click import of shipment manifests with automatic address geocoding, parcel categorization, and client identification. Import history and warnings are tracked for audit purposes.

Simulated Annealing Route Optimization

A 1,658-line routing engine that uses the simulated annealing metaheuristic to find near-optimal delivery sequences. Considers time windows, express priorities, vehicle constraints, driver shifts, depot locations, and real-time traffic conditions.

Live Traffic Integration

Real-time traffic data from Google Routes API with congestion level indicators (Normal/Slow/Heavy/Standstill), delay calculations, and traffic-aware ETA estimates.

Driver Companion App

A mobile-optimized PWA that provides turn-by-turn stop sequences, GPS tracking, stop action workflows (Arrive/Complete/Fail/Skip), signature capture, and trip summaries.

Dispatcher Live Map

Real-time Leaflet map showing all driver positions, next-stop ETAs, congestion indicators, and online/offline status with 60-second auto-refresh.

Data Centre

Complete archival system with PostgreSQL storage, searchable trip history, and CSV export for reporting and analysis.

Insights Engine

Intelligent alert system generating categorized recommendations for road conditions, fuel efficiency, punctuality, capacity utilization, and alternative route suggestions based on Pretoria corridor knowledge.

4. System Architecture & Tech Stack

Architecture Pattern

Full-stack monorepo with a React SPA frontend served by an Express.js backend, connected to PostgreSQL via Drizzle ORM. The system follows a local-first design — critical state is persisted in localStorage for instant access, with the database providing durability and multi-device sync.

Frontend Stack

- React 18 with TypeScript — Component-based UI with strict typing
 - Vite — Lightning-fast dev server with HMR and optimized production builds
 - shadcn/ui + Radix UI — 25+ accessible, customizable UI primitives
 - Tailwind CSS — Utility-first styling with dark/light theme support
 - TanStack React Query v5 — Server state management with caching
 - Wouter — Lightweight client-side routing (< 2KB)
 - Leaflet + react-leaflet — Interactive maps for driver tracking
 - Framer Motion — Fluid animations and page transitions
 - Recharts — Data visualization for metrics and analytics
 - Lucide React — Consistent icon set throughout the UI

Backend Stack

- Node.js with Express.js — REST API server
 - TypeScript — End-to-end type safety shared with frontend
 - PostgreSQL — Relational database with JSONB for flexible schemas
 - Drizzle ORM — Type-safe database queries with auto-migration
 - express-session + connect-pg-simple — Session-based dispatcher auth
 - jsonwebtoken (JWT) — Stateless driver app authentication
 - bcryptjs — Password hashing with 10 salt rounds
 - TSX — TypeScript execution without compilation step

External Services

- Google Places API (New) — Address autocomplete, geocoding, place details
 - Google Routes API — Traffic-aware routing, ETA calculations, polylines
 - OSRM (Open Source Routing Machine) — Free polyline generation (primary)
 - ShipLogic — CSV shipment manifest source system
 - CartoDB — Map tile provider for dark/light theme maps

DevOps & Infrastructure

- Replit — Cloud hosting, deployment, and development environment
 - Replit PostgreSQL — Managed database with automatic backups
 - PWA / Service Worker — Offline-capable driver app with installability
 - Vite PWA Plugin — Service worker generation and manifest injection

5. Feature Catalogue

5.1 Dispatcher Dashboard

The dispatcher dashboard is a multi-tab application providing complete operational control:

- Import Tab — CSV upload from ShipLogic with drag-and-drop, automatic parsing, address geocoding, duplicate detection, and import history tracking
- Operations Tab — Live schedule view with per-driver stop lists, drag-to-reorder, manual reassignment between drivers, waybill search across all tabs, and driver reorder request approval workflow
- Trips Tab — Per-driver route visualization with stop sequencing, ETA estimates, distance/time metrics, and trip cost breakdowns
- Traffic Tab — Live traffic dashboard with per-leg congestion indicators, fleet delay summaries, auto-refresh with countdowns, and road condition reporting for Gauteng corridors
- Metrics Tab — Delivery performance analytics with charts, fuel cost calculations, driver utilization, and efficiency comparisons
- Settings Tab — Fleet configuration (driver profiles, vehicles, depots, shifts, fuel price), driver account management for the companion app, user profile with avatar customization
- Data Centre Tab — Trip archival to PostgreSQL, searchable history with date/driver/waybill filters, CSV export, and detailed shipment-level data retention
- Handoffs Tab — Mid-journey parcel exchange management at predefined petrol station meeting points between drivers

5.2 Route Optimization Engine

The routing engine (1,658 lines of TypeScript) implements a simulated annealing optimization algorithm — a metaheuristic inspired by the metallurgical process of controlled cooling. The algorithm intelligently explores the solution space by:

- Evaluating millions of possible route permutations across all drivers simultaneously
- Accepting temporarily worse solutions early on (high temperature) to escape local optima
- Gradually reducing acceptance probability (cooling) to converge on near-optimal solutions
- Enforcing vehicle constraints — Refiloe's Toyota Vitz cannot carry 3-tier cakes, limited to 3 platter consignments and 6 parcels

- Respecting bread consignment rules — SWE001 restricted to i10 drivers (Vinny, Ashley, Clifford)
 - Honouring time windows for express/priority shipments with heavy penalties for violations
 - Applying Saturday delivery caps (12 morning / 8 afternoon per driver)
 - Generating handoff events at petrol station meeting points when inter-driver exchanges optimize the overall route
 - Considering real-time traffic data when available for departure-time-aware routing

5.3 Driver Companion App (PWA)

A separate mobile-first Progressive Web App accessible at `/driver/*` paths, designed to be installed on driver smartphones:

- Login screen with JWT authentication, 24-hour token expiry
- Dashboard showing today's stops (active, upcoming, completed) with colour-coded status
 - Interactive Leaflet map with blue pulsing dot for current location, numbered stop markers in driver's fleet colour
 - Stop action workflow: Arrive (GPS-stamped) !' Complete/Fail/Skip with completion modal
 - Completion modal captures: recipient name, signature pad (canvas-based), delivery notes
 - GPS tracking via `watchPosition` API with high accuracy, batched sends every 15 seconds
 - Stop reorder requests — drivers can propose alternative stop sequences with reasons, submitted for dispatcher approval
 - Pull-to-refresh with animated indicators and skeleton loading placeholders
 - Completed stops greyed out with reduced opacity; grey map markers
 - Trip summary page with delivery statistics when all stops are done
 - End Trip button that sets driver offline and logs out
 - Offline-capable with `localStorage` trip caching and graceful degradation
 - Separate PWA manifest (Delicate Driver) for independent home screen installation

5.4 Live Tracking & Traffic Integration

Real-time operational visibility through multiple interconnected systems:

- Dispatcher Live Map — Leaflet map panel showing all driver positions with coloured dots and name labels
- Traffic-aware ETAs from driver GPS to next stop using Google Routes API (TRAFFIC_AWARE routing preference)
 - Congestion level indicators: green (Normal), yellow (Slow), orange (Heavy), red (Standstill)

- Traffic delay callouts showing additional minutes due to congestion
 - Auto-refresh every 60 seconds via TanStack Query with collapsible panel
 - Offline drivers shown as grey dots with last-known position
 - Dark/light map tile themes matching the application theme
 - Auto-fit bounds to show all active drivers on the map
 - Auto-offline detection — drivers marked offline after 2 minutes without GPS update
 - Location history stored (every 3rd GPS update) for route replay and analysis

5.5 Data Centre & Archival

- One-click trip archival capturing complete shipment, driver, and metric data to PostgreSQL
 - Searchable archive with filters by date range, driver, waybill number, and client
 - CSV export of archived data for external reporting and analysis
 - Import history tracking with timestamps and warning logs
 - Audit log system recording significant operational events

5.6 Insights Engine

An intelligent advisory system that generates actionable recommendations:

- Road alerts — Construction, closures, and hazards on Pretoria corridors (N1, N4, N14, R21, etc.)
- Fuel recommendations — Cost optimization suggestions based on driver routes and vehicle consumption
- Punctuality analysis — Late delivery pattern detection and shift adjustment suggestions
 - Capacity alerts — Vehicle overloading warnings and load balancing recommendations
 - Alternative routes — Corridor-specific detour suggestions based on Pretoria local knowledge

6. API Cost Optimization

A major engineering focus was minimizing Google API costs while maintaining functionality. The system implements a multi-layered cost optimization strategy achieving an estimated 65-75% reduction in billable API calls compared to a naive implementation.

Centralized API Wrapper

All Google API calls are routed through a single centralized module (`server/lib/google-api.ts`) that provides:

- Per-type usage counters tracking total calls, cache hits, billable calls, and errors
 - In-flight request deduplication — concurrent identical requests share a single upstream call
- Request-level logging with timestamps for the usage dashboard
 - Protected admin endpoint (`/api/admin/api-usage`) for real-time cost monitoring

Server-Side TTL Caching

- Places Autocomplete cache: 5-minute TTL, max 200 entries — same search terms return cached results
 - Geocode cache: 24-hour TTL, max 500 entries — addresses rarely need re-geocoding
 - Place Details cache: 24-hour TTL, max 500 entries — place data is stable
- Traffic cache: 120-second TTL — balances freshness with cost
 - Dispatcher ETA cache: 300-second (5-minute) TTL — overview doesn't need sub-minute precision
 - Route polyline cache: 15-minute TTL — route geometry changes slowly

Client-Side Optimizations

- Autocomplete minimum input: 3 characters (prevents single/double letter API calls)
- Autocomplete debounce: 350ms (prevents rapid-fire calls during typing)
- Session tokens linking autocomplete !' details calls (Google bills as single session)
 - Batch traffic leg fingerprint diffing — only re-fetch legs whose coordinates changed (90s TTL)
 - Tab-visibility gating — traffic auto-refresh pauses when dispatcher views non-traffic tabs
 - Dispatcher map polling: 60 seconds (reduced from 15s)
 - Traffic auto-refresh: 10 minutes (reduced from 5

minutes)

Architectural Optimizations

- OSRM-first for polylines (free open-source routing) with Google as fallback only
 - Reduced field masks — no travelAdvisory or step-by-step directions requested
 - Driver ETAs use local calcDrive() math (zero Google API calls) — Haversine + city factor
 - Skip ETA recalculation when driver has moved less than 50 meters
 - Routes deduplicated on key includes departure time and routing preference to prevent cross-contamination
 - Strict coordinate validation (f

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7. Security & Authentication

Dispatcher Authentication

- Session-based auth with express-session and PostgreSQL-backed session store (connect-pg-simple)
- bcryptjs password hashing with 10 salt rounds
- 7-day session cookies with httpOnly flag, secure in production, sameSite: lax
- Protected API routes via requireDispatcherAuth middleware
 - User profile management with avatar colour picker, display name, email, and password change

Driver Authentication

- JWT-based stateless authentication using SESSION_SECRET
 - 24-hour token expiry requiring daily re-login
 - Auto-redirect on 401 responses with reactive auth state management
 - Driver accounts created and managed exclusively by dispatcher (no self-registration)

Data Protection

- GPS coordinates validated against South African boundaries before storage
 - Location update rate limiting: max 1 update per 10 seconds per driver
 - Location history auto-cleanup: records older than 30 days deleted daily
 - API keys stored as environment secrets, never exposed to frontend
 - Input validation and coordinate range checking on all geographic endpoints

8. Database Architecture

PostgreSQL database with 11 tables managed via Drizzle ORM with automatic schema synchronization:

users — Dispatcher accounts with profiles (display_name, email, role, avatar_color), bcrypt password hashing

session — PostgreSQL-backed session store for dispatcher authentication (connect-pg-simple, auto-managed)

projects — Active delivery projects with shipments, assignments, stop statuses, and driver sequences stored as JSONB

trip_archives — Archived trip data with complete shipment details, metrics, and driver assignments

csv_imports — CSV import history with timestamps, file metadata, and parsing warnings

audit_logs — System audit trail recording significant operational events with timestamps and details

app_settings — Key-value application settings for fleet configuration and system preferences

driver_accounts — Driver login accounts with real-time GPS position, fleet colour, vehicle plate, and vehicle type

driver_stop_events — GPS-stamped stop action events (arrive, complete, fail, skip) for proof of delivery

driver_location_history — Periodic GPS breadcrumb trail (every 3rd update) for route replay

driver_reorder_requests — Driver-initiated stop reorder requests with proposed order, reason, and approval status

9. ROI & Cost-Benefit Analysis

Time Savings

Task	Before	After	Savings
Manual Route Planning	45-60 min/day	2-3 min/day	~55 min/day saved
Driver Communication	30-40 min/day	5 min/day	~30 min/day saved
Shipment Data Entry	20-30 min/day	2 min (CSV import)	~25 min/day saved
End-of-Day Reporting	15-20 min/day	1-click archive	~18 min/day saved
Delivery Dispute Resolution	Variable	Digital proof on file	Significant reduction

Total Daily Time Savings: ~2 hours per day (~44 hours per month)

Fuel & Distance Savings

The simulated annealing route optimizer consistently produces routes 25-35% shorter than manual planning. For a 3-driver fleet averaging 150-200 km/day combined:

- Estimated distance reduction: 40-70 km/day across the fleet
 - At average fuel consumption of 7-9 L/100km and R22-24/litre fuel price:
 - Estimated fuel savings: R60-150 per day !' R 1,300-3,300 per month
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Optimization	Mechanism	Reduction
Server-side TTL caching	Autocomplete, geocode, place details, traffic	30-40%
OSRM-first polylines	Free open-source routing replaces paid Google	10-15%
Driver ETA using calcDrive	calls Eliminated Google API calls for driver ETAs	10-15%
Batch fingerprint diffing	entirely Only re-fetch changed legs in traffic batch	5-10%
Tab-visibility gating	Pause traffic refresh on non-traffic dispatcher tabs	3-5%
Longer polling intervals	Map 60s, traffic 10min, ETA cache 5min	5-8%

50m movement threshold

Skip ETA recalc when driver stationary

2-3%

Operational Improvements

- Real-time driver visibility eliminates "blind spots" in the delivery process
 - Digital proof of delivery (GPS stamps, signatures, recipient names) reduces disputes
 - Historical data archival enables data-driven route planning improvements over time
 - Vehicle constraint enforcement prevents loading errors and customer complaints
 - Satisfies

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- Insights

10. Project Statistics

Total Source Files 104 TypeScript/TSX/
CSS files

Frontend Code 19,394 lines (React +
TypeScript)

Backend Code 3,273 lines (Express +
TypeScript)

Shared Schema 412 lines (Drizzle ORM
models)

Routing Engine 1,658 lines (simulated
annealing optimizer)

CSS / Styling 238 lines (Tailwind +
custom properties)

Database Tables 11 PostgreSQL tables

NPM Dependencies 72 production + 23
development

UI Components 25+ shadcn/ui primitives

API Endpoints 30+ REST endpoints

Authentication Systems 2 (session-
based dispatcher + JWT driver)

Map Integrations 2 (Dispatcher live map
+ Driver route map)

External APIs 3 (Google Places, Google
Routes, OSRM)

PWA Manifests 2 (Dispatcher + Driver
companion)

Theme Support Dark and Light mode
with automatic detection

Key API Endpoints

POST /api/auth/login, /register, /logout — Dispatcher authentication
GET /api/auth/me, PATCH /api/auth/profile — Profile management
GET /api/projects, POST /api/projects — Project CRUD
POST /api/routes/traffic, /traffic/batch — Traffic data proxy
GET /api/routes/traffic/status — Traffic service health check
GET /api/dispatch/driver-locations — Live driver tracking
POST /api/driver/auth/login — Driver JWT login
POST /api/driver/location — GPS position updates
GET /api/driver/trip-sheet — Active trip data
POST /api/driver/stop/:key/arrive|complete|fail|skip — Stop actions
GET /api/driver/route-polyline — Route geometry
POST /api/driver/reorder-request — Stop reorder proposal
GET /api/admin/api-usage — API cost monitoring dashboard
GET /api/places/autocomplete, /geocode, /details — Places proxy

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Production-Ready Delivery Management Platform

Pretoria / Tshwane Metropolitan Area

Built with React, TypeScript, Express, PostgreSQL

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